



IILM UNIVERSITY, GREATER NOIDA

Department of Computer Science & Engineering

Python Full Stack Development

With Project — Internship Report

STUDENT

Mohd Kaif

PROGRAM

B.Tech CSE — Year 4 - Semester 7

ENROLLMENT NO.

CS-23411016

HOST ORGANIZATION

EduSkills Academy (AICTE)

Agenda



1. Introduction

Internship snapshot



2. Organization

AICTE–EduSkills program



3. Objectives

What the internship aimed to build



4. Tech Stack

Tools & technologies used



5. Architecture

How the system is structured



6. ER Diagram

Data model & relationships



7. Timeline

8-week training breakdown



8. Capstone Project

E-commerce platform build



9. Outcomes

Skills gained



10. Certification

Certificate & offer letter

Host Organization



EduSkills Academy

AICTE-recognized virtual internship provider partnering universities with industry-grade curricula.

8

WEEKS
DURATION

Apr-Jun
2026

PROGRAM
DATES

- **AICTE-EduSkills Virtual Internship**
A structured, remote 8-week program recognized under the AICTE internship initiative for engineering students.
- **Duration: April – June 2026**
Delivered in weekly modules combining self-paced learning, live mentoring sessions, and hands-on assignments.
- **Industry-Oriented Training**
Curriculum designed around real-world, full-stack development practices used in modern software teams.

Objectives

01



Master Core Technologies

Build strong command over Python, FastAPI, and React through structured, hands-on modules.

02



Build REST APIs

Design and implement clean, well-documented REST APIs following industry best practices.

03



Integrate SQLAlchemy

Model and connect relational data using SQLAlchemy ORM for reliable persistence.

04



Develop a Full-Stack App

Combine frontend, backend, and database layers into one deployable, working application.

Technology Stack

BACKEND



Python



FastAPI

FRONTEND



React

DATABASE



SQLAlchemy

TOOLING



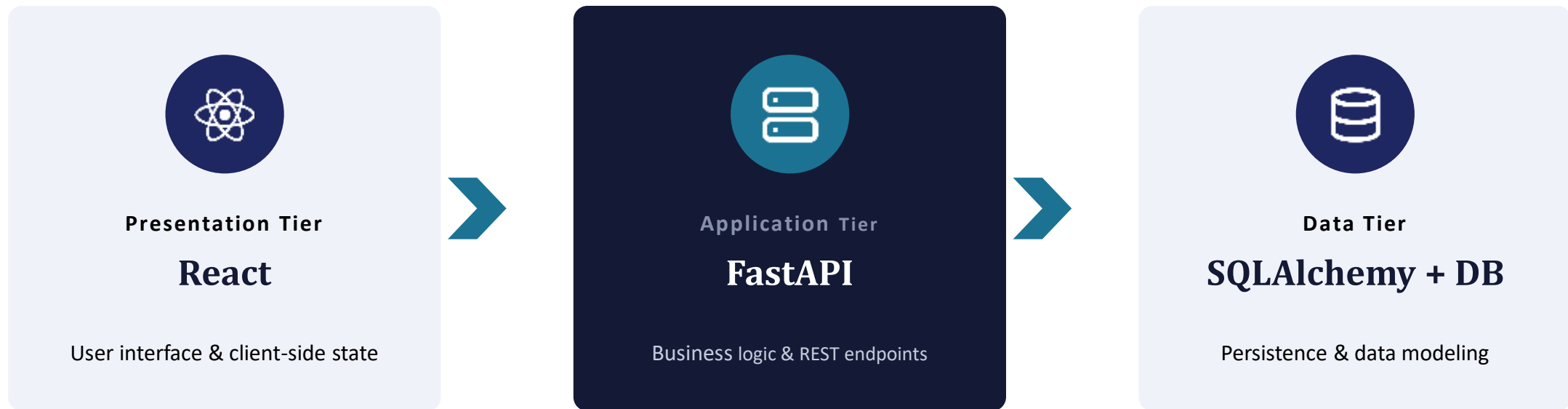
Docker



GitHub

System Architecture

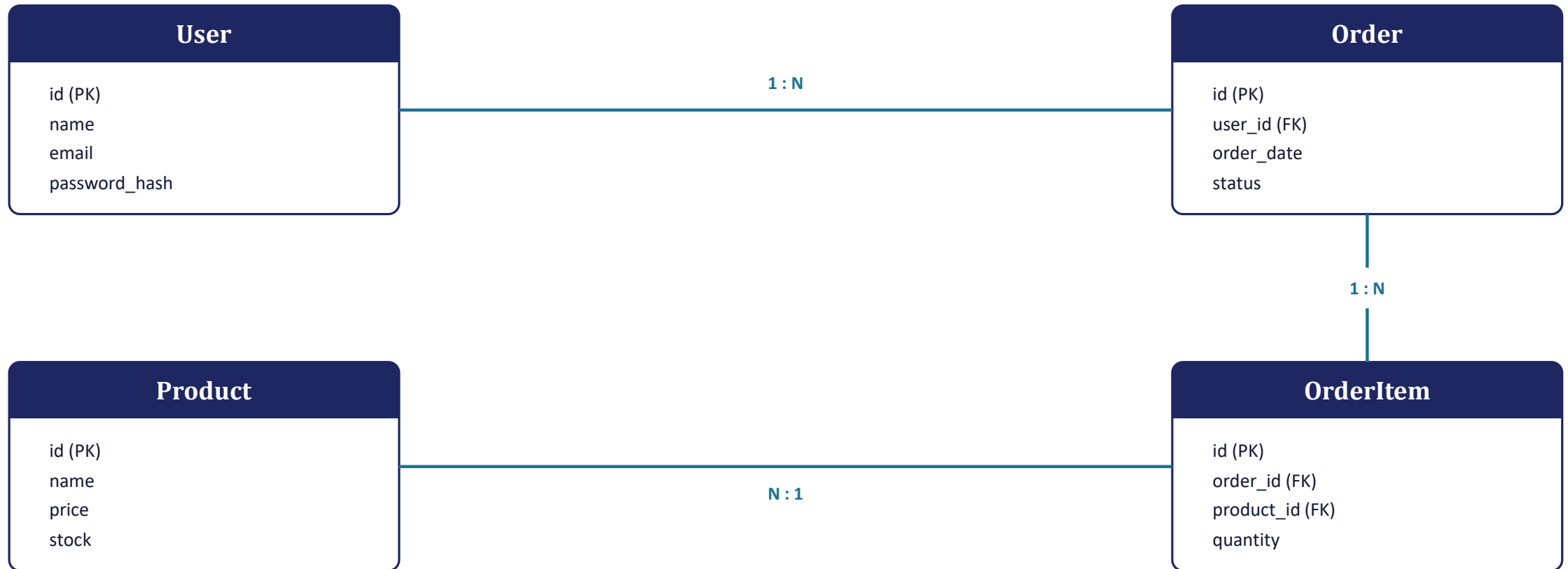
A three-tier architecture separates presentation, business logic, and data — each layer independently scalable and testable.



Performance Optimization: a TTL (Time-To-Live) Cache sits in the application tier to reduce repeated database hits and speed up frequent reads.

Entity-Relationship Diagram

Four core entities model the e-commerce domain, linked through well-defined relationships.



Week-Wise Timeline



Capstone Project



E-Commerce Platform

A working full-stack application built end-to-end during the internship, applying every layer of the tech stack in one deployable product.



Authentication

Secure user sign-up, login, and session handling.



CRUD Operations

Full create-read-update-delete flows for products and orders.



TTL Cache Lite

Lightweight time-based caching layer for faster responses.



Persistent Storage

Relational data managed through SQLAlchemy models.

Learning Outcomes



REST APIs

Designed and consumed RESTful services following clean API conventions.



Frontend-Backend Integration

Connected a React client to a FastAPI backend with reliable data flow.



Database Design

Modeled relational schemas and relationships using SQLAlchemy ORM.



Deployment Concepts

Learned containerization and version control fundamentals with Docker & GitHub.

Certificate & Offer Letter



Internship Successfully Completed

All eight weeks of the AICTE–EduSkills Python Full Stack Development program were completed, with the completion certificate and offer letter formally submitted.



Certificate Issued



Offer Letter Received

Thank You

Questions & Discussion

Mohd Kaif • CS-23411016 • B.Tech CSE, Semester 7th • IILM University